

Correlation

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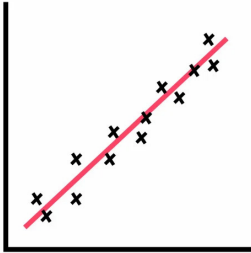
References

- Correlation is a statistical measure to assess whether two numerical variables are associated (linearly or non-linearly related)
- Most common correlation tests were:
 1. Pearson correlation test - part of parametric tests
 2. Spearman correlation test - part of non-parametric tests

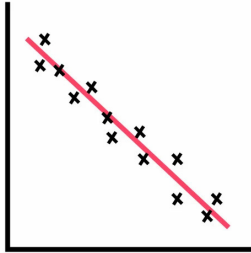
Pearson correlation test

- Aims: to describe the linear relationship between two numerical variables
- Pearson correlation test is conducted when:
 - Two numerical variable
 - At least one of the variables has normal distribution
- Assumptions:
 1. Random sample
 2. Normality - at least one numerical variables is approximately normally distributed
 3. No extreme outliers
- Examples:
 - Assessing the association between duration of exercise and systolic blood pressure
 - Assessing the association between length of hospital stay and medical cost

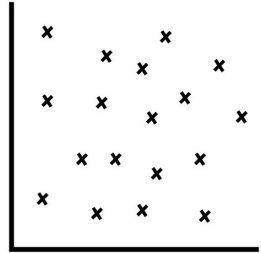
- Pearson correlation is range between -1 and 1
 - $r < 0$: negative correlation
 - $r = 0$: no correlation
 - $r > 0$: positive correlation



Positive
Correlation



Negative
Correlation



No
Correlation

- A more specific cut-offs (Taylor, 1990):
 - $0 < r < 0.35$: weak correlation
 - $0.36 < r < 0.67$: moderate correlation
 - $0.68 < r < 1$: high correlation

Pearson correlation test: Analysis

- Background: we want investigate the association between murder and assault cases in the US
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Interpretation
 4. Conclusion

- Step 1:
 - H_0 : There is no association between murder and assault cases
 - H_A : There is an association between murder and assault cases

Packages

```
library(knitr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

Read the data

```
dat <- read.csv("Data/correlation.csv")
```

```
# First 6 rows of the data
```

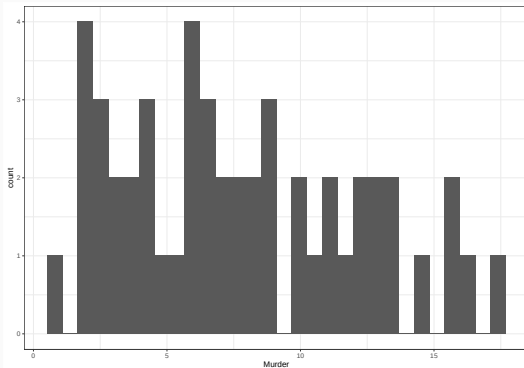
```
head(dat) %>% kable()
```

States	Murder	Assault	Rape
Alabama	13.2	236	21.2
Alaska	10.0	263	44.5
Arizona	8.1	294	31.0
Arkansas	8.8	190	19.5
California	9.0	276	40.6
Colorado	7.9	204	38.7

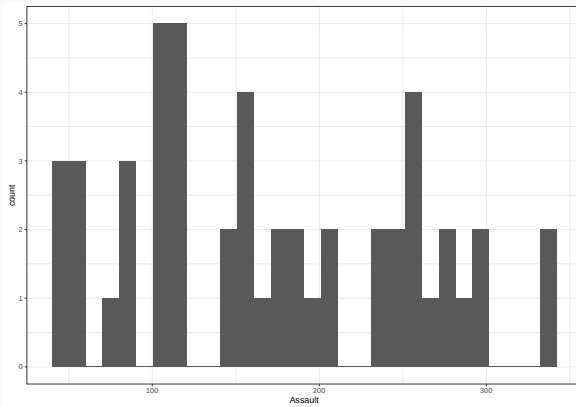
- Step 2: Assumptions:

1. Normality - let's assume both normally distributed

```
ggplot(dat, aes(Murder)) +  
  geom_histogram() +  
  theme_bw()
```



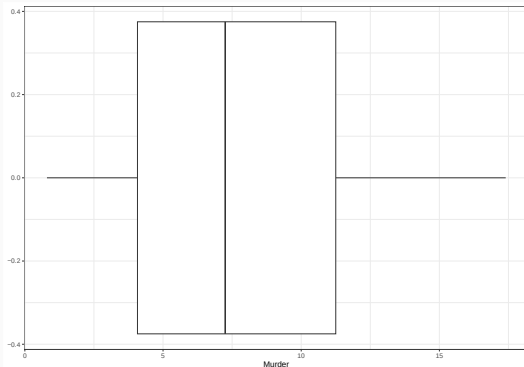
```
ggplot(dat, aes(Assault)) +  
  geom_histogram() +  
  theme_bw()
```



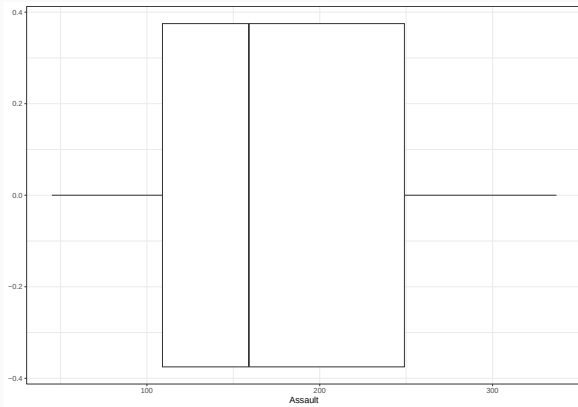
- Step 2: Assumptions:

- 2. No extreme outliers

```
ggplot(dat, aes(Murder)) +  
  geom_boxplot() +  
  theme_bw()
```



```
ggplot(dat, aes(Assault)) +  
  geom_boxplot() +  
  theme_bw()
```



- Step 3: Interpret:
 - p value < 0.05
 - 95% CI does not cross 0
 - There is a significant association between murder and assault cases

```
cor.test(dat$Murder, dat$Assault, method = "pearson")
```

Pearson's product-moment correlation

data: dat\$Murder and dat\$Assault

t = 9.2981, df = 48, p-value = 2.596e-12

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.6739512 0.8831110

sample estimates:

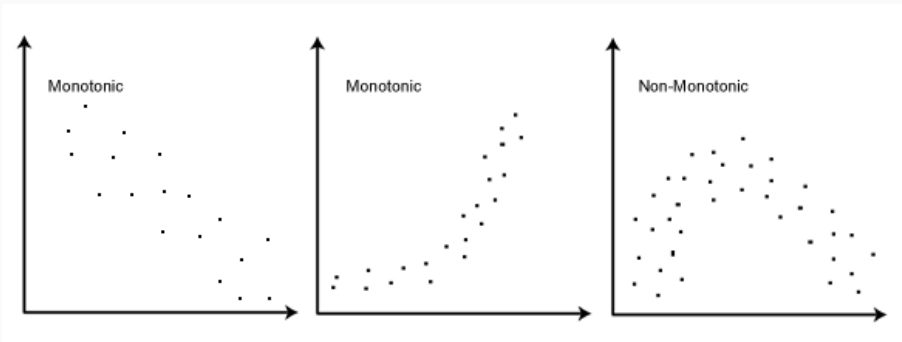
cor

0.8018733

- Step 4: Conclusion:
 - There is a significant correlation between murder and assault cases in the US
 - Pearson correlation coefficient, $r = 0.80$ indicates a positive high correlation (Taylor, 1990)

Spearman correlation test

- Aims: to describe the monotonic relationship between two numerical variables (not necessary a linear one - non-parametric test)
 - The rate of change may vary, but the direction of change remains consistent





Monotonically Increasing



Monotonically Decreasing



This is not a monotonic function

- Monotonically increasing - as the x variable increases the y variable never decreases
- Monotonically decreasing - as the x variable increases the y variable never increases
- Not monotonic - as the x variable increases the y variable sometimes decreases and sometimes increases.

- Spearman correlation test is conducted when:
 - Two numerical variable
 - Sample size is small ($n < 30$)
 - The numerical variables are not normally distributed
- Spearman correlation coefficient, p is ranged between -1 and 1
- We can use the same cut-offs(Taylor, 1990):
 - $0 < p < 0.35$: weak correlation
 - $0.36 < p < 0.67$: moderate correlation
 - $0.68 < p < 1$: high correlation

- Assumptions:
 1. Random sample
- Examples:
 - Assessing the association between duration of exercise and systolic blood pressure
 - Assessing the association between length of hospital stay and medical cost

Spearman correlation test: Analysis

- Background: we want investigate the association between rape and assault cases in the US
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Interpretation
 4. Conclusion

- Step 1:
 - H_0 : There is no association between rape and assault cases
 - H_A : There is an association between rape and assault cases

Packages

```
library(knitr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

Read the data

```
dat <- read.csv("Data/correlation.csv")
```

```
# First 6 rows of the data
```

```
head(dat) %>% kable()
```

States	Murder	Assault	Rape
Alabama	13.2	236	21.2
Alaska	10.0	263	44.5
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Arkansas	8.8	190	19.5
California	9.0	276	40.6
Colorado	7.9	204	38.7

- Step 2: Assumptions: check theoretical assumption
- Step 3: Interpret:
 - $p \text{ value} < 0.05$
 - There is a significant association between rape and assault cases

```
cor.test(dat$Rape, dat$Assault, method = "spearman")
```

Spearman's rank correlation rho

data: dat\$Rape and dat\$Assault

S = 5948.3, p-value = 5.689e-09

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.7143681

- Step 4: Conclusion:
 - There is a significant correlation between rape and assault cases in the US
 - Spearman rho, $r = 0.71$ indicates a positive high correlation (Taylor, 1990)

Taylor, R. (1990). Interpretation of the correlation coefficient: A basic review. *Journal of Diagnostic Medical Sonography*, 6(1), 35–39.